

# Equatorial wave theory, methodology for identifying equatorial waves, and dataset

## WCSSP WESTPAC Project

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### 1. Basic equatorial wave theory

Equatorially trapped waves are obtained as solutions to the adiabatic, frictionless equations of motion on an equatorial  $\beta$ -plane linearized about a state of rest and separation of the vertical structure from that in the horizontal (Matsuno 1966; Gill 1980). The horizontal and temporal behavior of horizontal winds ( $u, v$ ) and geopotential height ( $Z$ ) satisfy the linearized shallow water equations with gravity wave speed  $c$ , the separation constant from the vertical structure equation that can also satisfy relevant surface and upper boundary conditions. This is possible only for discrete values of the separation constant,  $c_e$ . In an atmosphere with a constant buoyancy frequency  $N$  and a tropopause at height  $H$  with a rigid lid upper boundary condition, the separation coefficient is given by  $c_e = NH/m\pi$ , where  $m$  is the vertical mode number. The vertical modes are sinusoidal in  $z$ , with corrections for the density variation with height:  $1/\rho_0 \exp(z/2H) \exp(imz)$ , where  $\rho_0$  is the density at the surface.

For the horizontal equations, the representation of  $u, v$  and  $Z$  fields are of the form:

$$\{u, v, Z\} = \{U(y), V(y), Z(y)\} \exp[i(kx - \omega t)]. \quad (1)$$

where  $k$  is the zonal wavenumber and  $\omega$  is the frequency. The equatorial wave solutions are most easily formulated in terms of new variables,  $q, r$  and  $v$  (Gill 1980) where

$$q = \frac{gZ}{c_e} + u, \quad r = \frac{gZ}{c_e} - u \quad (2)$$

There is the Kelvin wave solution with zero  $v$  and  $\omega = k/c_e$ , and there are solutions with non-zero  $v$  with the dispersion relation

$$\frac{\omega^2}{c_e \beta} - c_e \frac{k}{\omega} - \frac{c_e}{\beta} k^2 = (2n + 1), \quad \text{for } n = 0, 1, 2, \dots \quad (3)$$

where  $n$  is the meridional mode number and  $\beta$  is  $\partial f / \partial y$ . Since the Kelvin wave satisfies this relation with  $n = -1$ , this notation is conventionally used to label it. The Kelvin wave is eastward moving. The  $n=0$  mode is the mixed Rossby-gravity (MRG) wave which has both eastward (EMRG) and westward-moving (WMRG) solutions. For  $n=1$  and higher there are westward-moving equatorial Rossby waves and both eastward and westward-moving gravity wave solutions.

The meridional ( $y$ ) structures of equatorial waves can be described by parabolic cylinder functions:

$$D_n\left(\frac{y}{y_0}\right) = \exp\left[-\frac{1}{4}\left(\frac{y}{y_0}\right)^2\right] P_n\left(\frac{y}{\sqrt{2}y_0}\right), \quad (4)$$

where  $y_0 = \left(\frac{c_e}{2\beta}\right)^{1/2}, \quad (5)$

is the meridional scale,  $P_n$  is proportional to a Hermite polynomial of order  $n$ , and the waves are trapped at the equator on a scale  $y_t = \sqrt{2} y_0$ .

Guided by basic equatorial wave theory, the parabolic cylinder function expansions are organized and described as follows:

$$\begin{aligned}
 q &= q_0 D_0 + q_1 D_1 + \sum_{n=1}^{n=\infty} q_{n+1} D_{n+1} \\
 v &= 0 + v_0 D_0 + \sum_{n=1}^{n=\infty} v_n D_n \\
 r &= 0 + 0 + \sum_{n=1}^{n=\infty} r_{n-1} D_{n-1}
 \end{aligned} \tag{6}$$

$\begin{matrix} \uparrow & \uparrow & \uparrow \\ n=-1 & n=0 & n=1, 2, \dots \end{matrix}$

These functions form a complete and orthogonal basis, and the projections in Eq.(6) are quite general, with  $q_0 D_0$  describing the Kelvin wave,  $q_1 D_1$  and  $v_0 D_0$  describing  $n=0$  MRG waves, and  $q_{n+1} D_{n+1}$ ,  $v_n D_n$  and  $r_{n-1} D_{n-1}$  describing  $n \geq 1$  equatorial Rossby waves or gravity waves. The theoretical horizontal structures of some of gravest equatorial waves are shown in Fig.1.

## 2 Methodology to identify equatorial waves from observations and simulations

Many previous studies used a method to identify equatorial waves by isolating disturbance in zonal wavenumber and frequency domains confined by theoretical dispersion curves for the waves (e.g., Wheeler and Kiladis 1999). However, although the separation of the vertical and horizontal structures is possible for a resting atmosphere, in general the separation of variables for observed atmospheric disturbances is not possible due to atmosphere being far from resting, shear in the zonal flow and the lack of rigid lid so that analysis in terms of vertical modes and horizontal wave structures is not strictly valid. Especially strong basic flow in the tropics would cause waves to be Doppler shifted (e.g., intrinsic westward-moving WMRG and R1 waves can be Doppler-shifted to move eastward in the upper tropospheric westerly ducts as shown in Yang and Hoskins 2016). Hence in Yang et al. (2003) a methodology to identify equatorially-trapped waves in observational data was developed. In the methodology, it was not assumed that the linear adiabatic theory for equatorial waves on a resting atmosphere is directly applicable, in particular, the dispersion relation and vertical structure are not imposed. Potential equatorial waves are identified by projecting the dynamical fields in the tropics at each pressure level onto the different equatorial wave modes using their horizontal structures described by parabolic cylinder functions in  $y$  and sinusoidal variation in  $x$ .

The analysis method is described as follows:

- 1) Formulate equatorial wave solutions in terms of  $v$  and two new variables:

$$q = \frac{gZ}{c_e} + u, \quad r = \frac{gZ}{c_e} - u, \tag{7}$$

where  $c_e$  is related to the trapping scale  $y_0 \left(\frac{c_e}{2\beta}\right)^{1/2}$  predetermined by data fit.

2) Filter data in the wavenumber and frequency domain.

Before projection onto the equatorial structure, the  $v$ ,  $q$  and  $r$  in the global tropical belt ( $24^\circ\text{N}$  and  $24^\circ\text{S}$ ) can be separated into eastward and westward-moving components using a space-time spectral analysis which transforms data from the  $x$ - $t$  domain into the  $k$ - $\omega$  domain by performing 2-D FFT in the zonal and time direction (Hayashi 1982). To avoid aliasing filtering procedure, two dominant periodic cycles, the climatological annual cycle and diurnal cycle, have been removed from the raw data before filtering. The data is then filtered using a specified spectral domain.

Although filtering data in a wavenumber-frequency domain can effectively separate waves into eastward and westward-moving modes, this is not suitable for real-time application or forecast data, because of the need for future information.

3) Project filtered data onto horizontal structures of equatorial waves.

The Fourier coefficients (e.g.,  $V(y)$ , for each  $k$  and  $\omega$ ) of eastward or westward-moving  $v$ ,  $q$  and  $r$  are separately projected onto the meridional structures of the equatorial waves as described by the parabolic cylinder functions in Eq. (6) to obtain the equatorial wave modes at each level. This allows the data itself to show the vertical structures, without assuming a vertical structure, and obtain amplitudes for  $v$ ,  $q$  and  $r$  separately without assuming that the amplitude coefficients for the three variables are related via dispersion relations.

In the projection, it is necessary to first specify the meridional scale  $y_0$  (or equivalently the trapping scale  $y_t$ ) and hence the speed  $c_e$ , so that  $q$  and  $r$  can be formed from  $u$  and  $Z$  according to Eq.(2). As in previous studies (e.g., Yang et al. 2003, 2007a,b, 2012),  $y_0 = 6^\circ$  (the trapping scale  $y_t \approx 8.4^\circ$ ) is used. The value of  $y_0$  was determined from a best fit to the data, although it was found that the analysis is not sensitive to the particular value of chosen  $y_0$ . From Eq. (5),  $c_e$  is determined to be about  $20 \text{ ms}^{-1}$ . This  $c_e$  is used only to create the new dependent variables  $q$  and  $r$  from  $u$  and  $Z$  and later to reverse the variable transform.

It should be noted that for a given meridional mode number  $n$  ( $n \geq 0$ ), because the meridional structures are the same for EMRG and WMRG, and the same for Rossby wave and gravity waves, the projection itself cannot distinguish them. However, the different wave types can be easily seen from their horizontal wind structures. As shown in Fig.1, for rotational waves, such as WMRG and R1, their zonal wind and meridional wind divergence ( $du/dx$  and  $dv/dy$ ) have an opposite sign, but for gravity wave type, such as EMRG and  $n=1$  gravity waves, their  $du/dx$  and  $dv/dy$  have a same sign. Based on previous studies the  $n \geq 0$  modes for period longer than two days are dominated by WMRG and R1 structures, hence we label the projected  $n=0$  and  $n=1$  modes as WMRG and R1 waves.

4) Transform the Fourier coefficients for each wave mode back into physical space and deduce  $u$  and  $Z$  for each wave mode from projected  $q$  and  $r$ .

This level-by-level projection technique has been successfully employed in a wide range of studies to investigate the horizontal and vertical structures of convectively coupled equatorial waves both in observations and models (e.g., Yang et al. 2007a,b and Yang et al. 2009), the

variation of equatorial waves in different phases of QBO (Yang et al. 2011, 2012), the influence of ENSO on equatorial waves and tropical convection (Yang and Hoskins 2013, 2016), connection between African Easterly wave and equatorial waves (Yang et al. 2018), linking extreme precipitation in Southeast Asia to equatorial waves (Ferrett et al. 2020), and identification of equatorial waves in rea-time forecast (Yang et al. 2021).

### 3 Historical dataset

A historical dataset for equatorial wave using ERA5 data has been produced for the WESTPAC project. The dataset spans 39 years from 1980 to 2018. It includes equatorial waves with zonal wavenumber  $k=2-40$  and period=2-30days. The Kelvin wave is filtered in the eastward-moving domain, and WMRG, R1 and R2 waves are filtered in westward-domain. They are on 6 hourly and 1deg resolutions at 28 pressure levels: 1000, 975, 950, 925, 900, 875, 850, 825, 800, 775, 750, 700, 650, 600, 550, 500, 450, 400, 350, 300, 250, 225, 200, 175, 150, 125, 100 and 70 hPa.

To reduce the file size, the dataset contains coefficients  $q_n$ ,  $v_n$  and  $r_n$  of projected  $q$ ,  $v$  and  $r$ . these coefficients are the function of longitude, pressure level and time. Since wave meridional structures are simply described by parabolic cylinder functions  $D_n(y)$ , wave variables  $v$ ,  $u$  and  $Z$  can be readily deduced from these coefficients and associated  $D_n$  (based on Eqs. 6 and 7). In doing so the file size is reduced by 49 times (49 latitude grids).

The solutions of  $u$ ,  $Z$  and  $v$  are summarised in Table 1.

Table 1  $v$ ,  $u$ ,  $Z$  solutions of equatorial waves, where  $g=9.8 \text{ m s}^{-2}$ ,  $c_e=20.4753 \text{ m s}^{-1}$

| Wave mode          | Meridional wind $v$ | Zonal wind $u$                    | Geopotential height $Z$                         |
|--------------------|---------------------|-----------------------------------|-------------------------------------------------|
| $n=-1$ Kelvin wave | $v = 0$             | $u = \frac{q_0 D_0}{2}$           | $Z = \frac{c_e}{g} \frac{q_0 D_0}{2}$           |
| $n=0$ WMRG wave    | $v = v_0 D_0$       | $u = \frac{q_1 D_1}{2}$           | $Z = \frac{c_e}{g} \frac{q_1 D_1}{2}$           |
| $n=1$ R1 wave      | $v = v_1 D_1$       | $u = \frac{q_2 D_2 - r_0 D_0}{2}$ | $Z = \frac{c_e}{g} \frac{q_2 D_2 + r_0 D_0}{2}$ |
| $n=2$ R2 wave      | $v = v_2 D_2$       | $u = \frac{q_3 D_3 - r_1 D_1}{2}$ | $Z = \frac{c_e}{g} \frac{q_3 D_3 + r_1 D_1}{2}$ |

The files contain  $q_n$ ,  $v_n$  and  $r_n$  for one calendar year and are named following

qvr\_coefficient\_YYYY\_k2-40\_p2-30\_28plev\_era5.nc

where

YYYY refers to the calendar year

k2-40\_p2-30 refers to the zonal wave number ( $k$ ) and period ( $p$ ) in days of the spectral filter

The data are held on JASMIN in:

/gws/nopw/j04/klingaman/gyang/era5/wave\_data/k240\_p230/

The file for parabolic cylinder functions (Dn.nc) is also in the above path.

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### **Terms of Use:**

This data is available for re-use under the WCSSP Southeast Asia project and more widely. When using the data in publications or other outputs please acknowledge the source of the data with the following statement:

“The equatorial wave dataset was produced by Dr Gui-Ying Yang of the National Centre for Atmospheric Science, University of Reading, supported by the Met Office Weather and Climate Science for Service Partnership (WCSSP) Southeast Asia as part of the Newton Fund. ”

and by including an appropriate reference to Yang et al. (2003) in the description of the data in the body of the text.

For re-use within the project, especially on the first occasion of your use, please consider including Gui-Ying Yang as a co-author of the publication

Please do not redistribute this data without this accompanying documentation.

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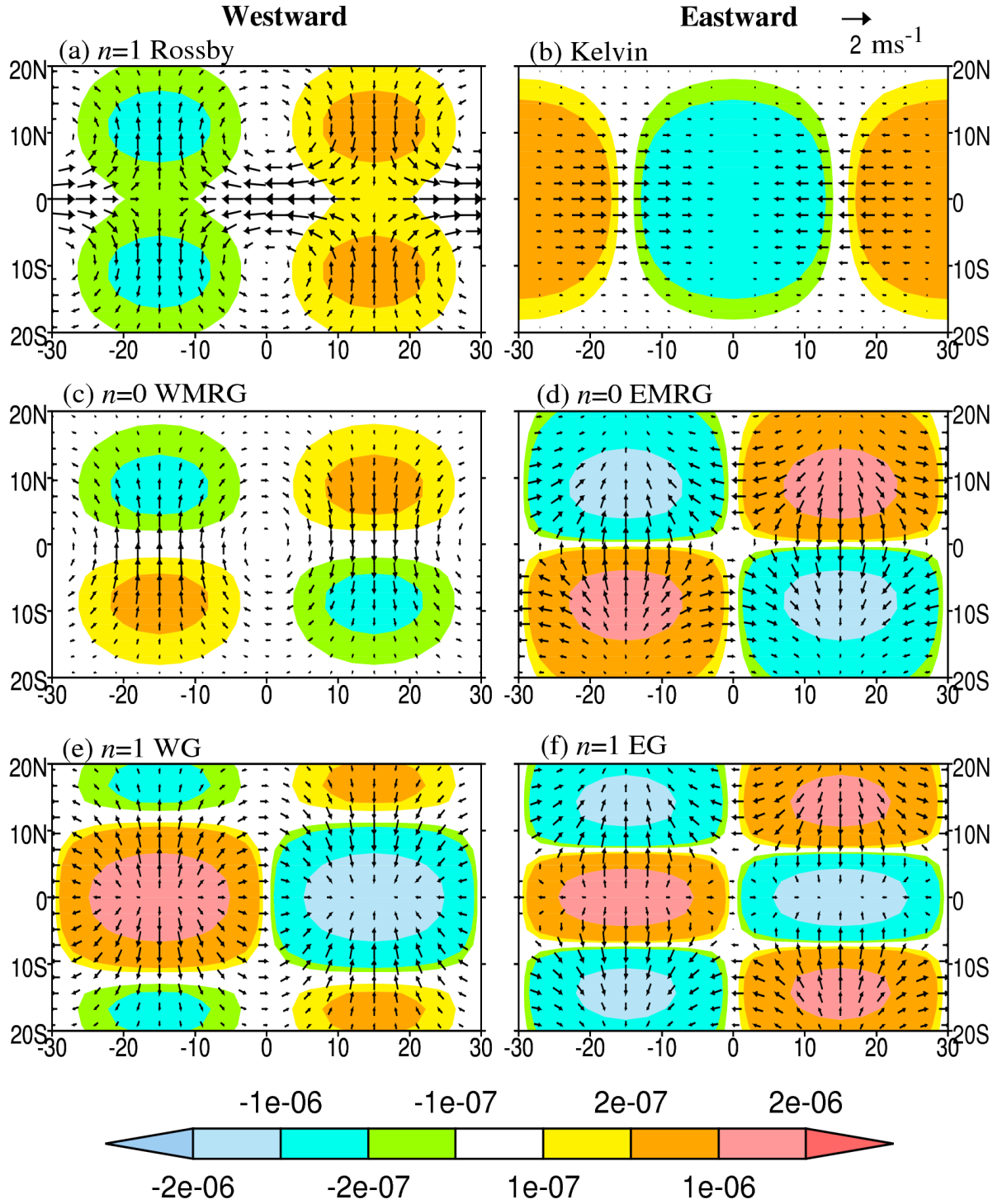


Figure 1. The theoretical horizontal structures of some of the gravest equatorial wave modes, the Kelvin wave, the  $n=0$  westward-moving mixed Rossby-gravity (WMRG) and eastward-moving mixed Rossby-gravity (EMRG), the  $n=1$  westward-moving Rossby (R1) and gravity (WG1) and eastward-moving gravity (EG1) waves. Vectors indicate horizontal wind and colors divergence. The maximum zonal wind in the Kelvin wave has been taken to be  $1 \text{ m s}^{-1}$ , and the maximum meridional winds in the  $n=0$  MRG waves to be  $1 \text{ m s}^{-1}$  and in the  $n=1$  waves  $0.86 \text{ m s}^{-1}$ . The trapping scale has been taken to be  $6^\circ$  and the zonal wave number  $k=6$ . Units are  $\text{m s}^{-1}$  for wind and  $\text{s}^{-1}$  for divergence.